IP ADDRESSING AND SUBNETTING

Assignment

Name: _____ Class: _____

Date: _____

NETWORK

TASK 1

QUESTION

List the IP address classes (A, B, C, D, E) and give one real-world example of a device or service that might use each class. Write your answers in a table format.

ANSWER

IP Class	First Octet Range	Example Use
Class A	1–126 in the first octet (traditional classful range)	A very large private or enterprise network; for example, a large organization using a private 10.0.0.0/8 network.
Class B	128–191 in the first octet	A medium-to-large organization or campus network; for example, a college network using private addresses from 172.16.0.0/16.
Class C	192–223 in the first octet	A small LAN; for example, a home router assigning addresses such as 192.168.1.10 to a laptop.
Class D	224–239 in the first octet	Multicast services, such as a network service sending one stream to a subscribed group of devices.
Class E	240–255 in the first octet	Reserved/experimental address space; it is not normally assigned to everyday host devices.

Note: Classful addressing is a traditional system. Modern networks generally use CIDR notation, but class ranges remain useful for learning.

NETWORK

TASK 2

QUESTION

Given IP address 192.168.10.45 with subnet mask 255.255.255.0, calculate the network address, broadcast address, and valid host range. Show working steps.

ANSWER

Given: IP address = 192.168.10.45; subnet mask = 255.255.255.0 (/24)

Step 1 — Network address: With a /24 mask, the first three octets are the network portion and the last octet is the host portion. Set the host portion to 0. **Network address = 192.168.10.0**

Step 2 — Broadcast address: Set all host bits (the last octet) to 1, giving 255 in the last octet. **Broadcast address = 192.168.10.255**

Step 3 — Valid host range: The first usable address is one above the network address, and the last usable address is one below the broadcast address. **Valid host range = 192.168.10.1 to 192.168.10.254**

Usable host addresses = 2■ – 2 = **254**.

NETWORK

TASK 3

QUESTION

Create a subnetting plan for a food delivery startup needing four separate subnets: Delivery Partners, Restaurants, Customers, and Admins. Use 10.0.0.0/24.

ANSWER

A /24 network has 256 addresses. Four equal subnets require borrowing 2 host bits: /24 + 2 = /26. Each /26 subnet has 64 total addresses and 62 usable host addresses.

Group	Network	Subnet Mask	Valid Hosts	Broadcast
Delivery Partners	10.0.0.0/26	255.255.255.192	10.0.0.1 – 10.0.0.62	10.0.0.63
Restaurants	10.0.0.64/26	255.255.255.192	10.0.0.65 – 10.0.0.126	10.0.0.127
Customers	10.0.0.128/26	255.255.255.192	10.0.0.129 – 10.0.0.190	10.0.0.191
Admins	10.0.0.192/26	255.255.255.192	10.0.0.193 – 10.0.0.254	10.0.0.255

Separate subnets help organize traffic and apply different access/security rules.

NETWORK

TASK 4

QUESTION

A college Wi-Fi network uses 172.16.0.0/16 and needs at least 20 subnets. Calculate the new subnet mask and hosts per subnet, showing steps.

ANSWER

Step 1 — Bits to borrow: 2■ = 16 subnets (not enough); 2■ = 32 (enough). Borrow 5 bits.

Step 2 — New prefix: Original /16 + 5 borrowed bits = **/21**.

Step 3 — Subnet mask: /21 = **255.255.248.0**.

Step 4 — Usable hosts: Host bits remaining = 32 – 21 = 11. Total addresses = 2^{11} = 2,048. Usable hosts = 2^{11} – 2 = **2,046**.

Final answer: Subnet mask 255.255.248.0 (/21), 32 subnets, and 2,046 usable host addresses per subnet.

NETWORK

TASK 5

QUESTION

Explain why IP address 192.168.1.300 will not connect and give examples of valid addresses in that subnet.

ANSWER

The address 192.168.1.300 is invalid because each IPv4 octet must be between 0 and 255. The last octet, 300, is too large.

A valid host address in the 192.168.1.0/24 subnet has the form 192.168.1.x, where x is normally 1–254. The network address is 192.168.1.0 and the broadcast address is 192.168.1.255.

Examples: **192.168.1.25** or **192.168.1.100**, provided the address is not already in use and matches the network configuration.

NETWORK